Healthcare Benefits Navigator: A RAG-Based System for Personalized Health Insurance Benefit Discovery and Recommendations

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Abstract

Health insurance selection is a complex, high-stakes decision that many consumers struggle to navigate effectively. In Massachusetts, the Health Connector offers over 30 plan variants across 8 licensed carriers and 5 metal tiers for plan year 2025, creating a decision space that is difficult to navigate without expert guidance. This paper presents the Massachusetts Health Insurance Navigator, a conversational AI system that enables residents to understand, compare, and select health insurance plans through natural language interaction. The system is built on a Retrieval-Augmented Generation (RAG) architecture combining three components: dense semantic retrieval using Sentence-BERT (all-MiniLM-L6-v2), learning to rank re-ranking using XGBoost, and natural language answer generation using Mistral AI (mistral-small-latest). Plan data was sourced from the CMS State-Based Exchange Public Use Files (PUF) 2025 and the MA Health Connector ConnectorCare Overview, encompassing 30 health plans and 7 ConnectorCare subsidy tiers. The system is deployed as a Streamlit web application featuring real-time plan filtering, structured plan comparison tables, and a star-rating feedback mechanism for continuous improvement. The system is evaluated using a combination of retrieval metrics (Hit Rate, MRR, NDCG), re-ranking metrics (Precision@K), and LLM quality metrics under an LLM-as-judge framework. The final system is deployed as a publicly accessible Streamlit web application, providing Massachusetts residents with an intuitive, reliable, and explainable health insurance navigation tool. This work demonstrates the viability of domain-constrained RAG pipelines for structured, tabular knowledge bases in regulated consumer information domains, and contributes a hybrid retrieval approach combining neural embeddings with structured metadata re-ranking.

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Introduction

Background and Problem Statement

Massachusetts established near-universal health coverage through the 2006 Health Care Reform Act, predating the federal ACA of 2010. The MA Health Connector marketplace offers plans from 8 licensed carriers under a strict 2:1 age rating cap (versus the federal 3:1 cap) and full gender parity under ACA Section 2701. The ConnectorCare program provides subsidized plans for households earning 0 to 500% of the Federal Poverty Level (FPL) across 7 plan types with $0 deductibles, adding further complexity for consumers.

Despite these protections, selecting a plan remains cognitively demanding. Consumers must evaluate 21 benefit categories, age-adjusted premiums, copays, out-of-pocket maximums, prescription tiers, and subsidy eligibility simultaneously. Existing tools present static tables with no natural language support or personalized filtering. Standalone LLMs are unsuitable due to knowledge cutoffs and hallucination risk in regulated domains.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), addresses these issues by grounding LLM responses in a retrieved external knowledge base at inference time. This project applies RAG to MA health insurance, enabling residents to ask natural language questions and receive accurate, context-grounded answers based on official 2025 plan data.

Objectives and Scope

The system aims to: (1) construct a structured 2025 MA plan knowledge base from CMS PUF and ConnectorCare data; (2) build a three-stage RAG pipeline;(3) deploy an accessible web application with age, tier, carrier, and ConnectorCare filtering; and (4) evaluate all pipeline stages with standard IR and NLP metrics.

Scope is limited to 30 individual MA plans from 8 carriers across Bronze, Silver, Gold, and Platinum tiers, and 7 ConnectorCare subsidy types (0 to 500% FPL) for plan year 2025. Medicare, Medicaid, dental-only plans, and employer-sponsored insurance are out of scope.

Technical Approach and System Architecture

System Architecture

The system follows a five-stage pipeline shown in Figure 2.1.

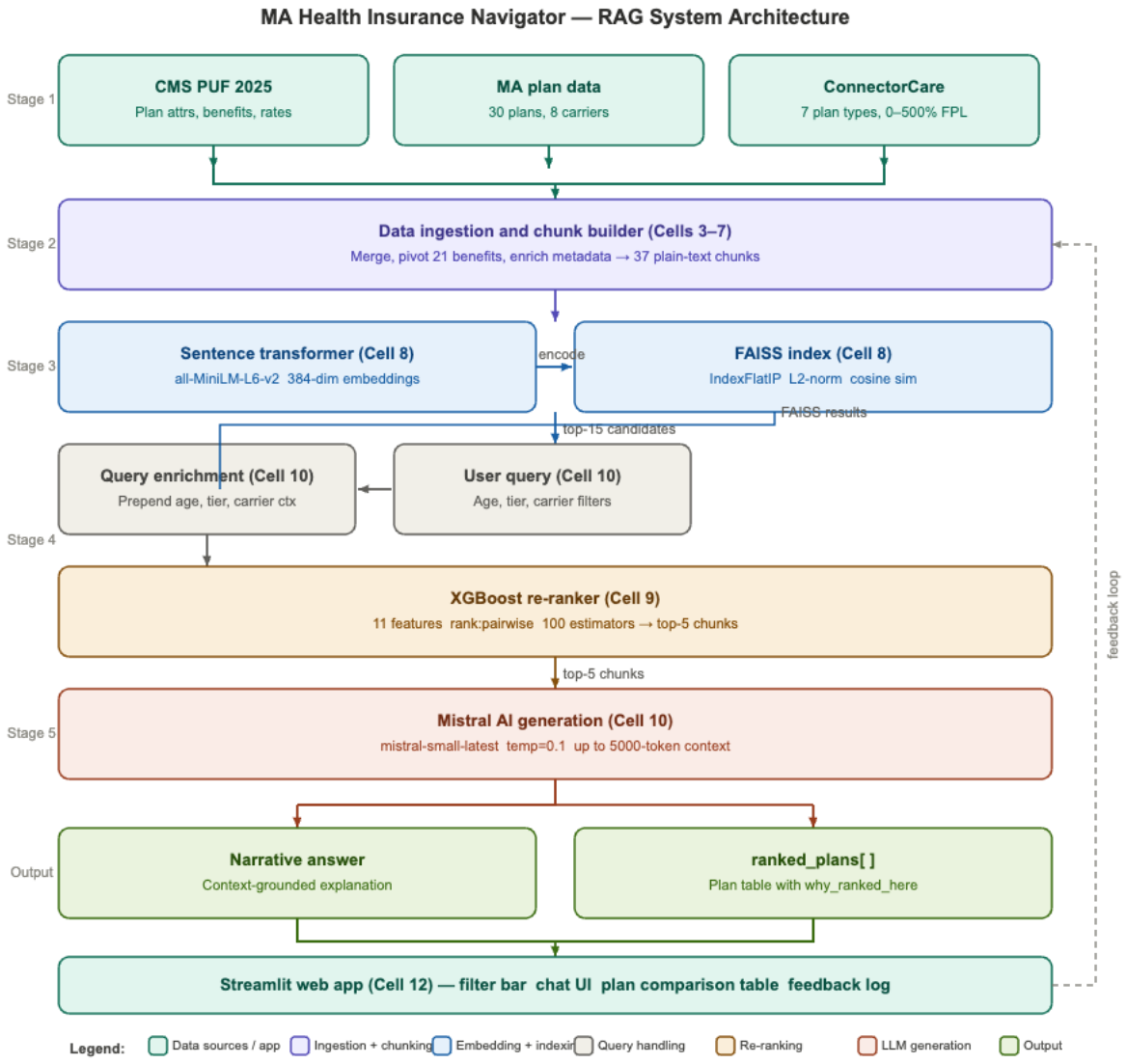

Figure 2.1: End-to-end RAG pipeline: data ingested from CMS PUF and ConnectorCare, converted to text chunks, indexed with FAISS, re-ranked with XGBoost, and passed to Mistral for structured JSON generation.

The five stages are: (1) **Data ingestion**: CMS PUF CSVs loaded, filtered to MA, pivoted across 21 benefit categories, enriched with gender parity and insulin cap metadata (Cells 3 to 5). (2) **Chunk construction**: each plan row and ConnectorCare entry serialized to a structured plain-text chunk, yielding 37 chunks total (Cell 7). (3) **Semantic indexing**: chunks encoded with `all-MiniLM-L6-v2` (384-dim embeddings), L2-normalized, stored in FAISS `IndexFlatIP` (Cell 8). (4) **Re-ranking**: XGBoost re-scores top-15 FAISS candidates using 11 features, returns top-5 (Cell 9).(5) **Generation**: Mistral (`temperature=0.1`) returns a JSON object with a narrative answer and ranked plan comparison table (Cell 10).

Data Preparation

Plan Dataset

The knowledge base covers 30 plans from 8 carriers (Table 2.1). Figure 2.2 shows distributions by carrier, metal tier, and plan type. Silver tier has the highest count (11 plans); 73.3% of plans are HMO.

Table 2.1: MA Licensed Carriers and Plan Counts (2025)

Carrier	Plans	Types
Blue Cross Blue Shield MA	7	HMO, PPO
Harvard Pilgrim	4	HMO, PPO
Tufts Health	4	HMO, PPO
UnitedHealthcare	4	HMO, PPO
Fallon Health	3	HMO
Health New England	3	HMO
Mass General Brigham	3	HMO
WellSense	2	HMO
Total	**30**	

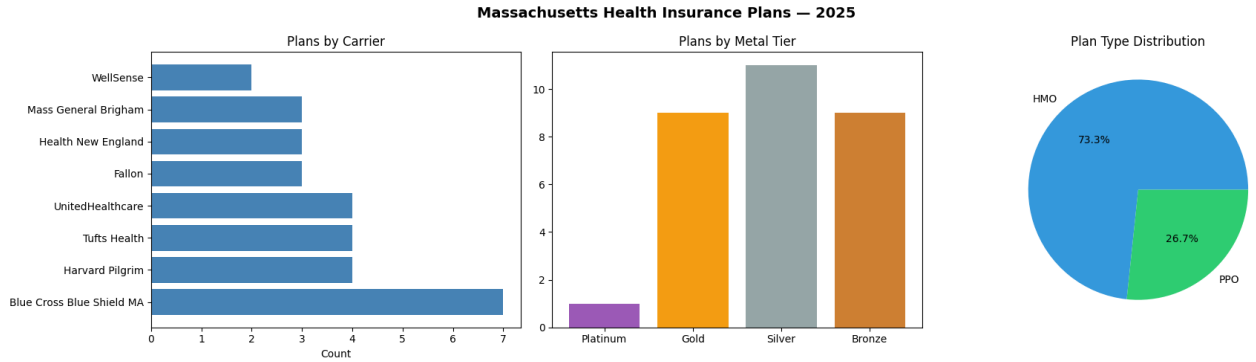

Figure 2.2: EDA (Cell 6): plan counts by carrier (left), metal tier (centre), and plan type (right). BCBS leads with 7 plans; Silver has 11 plans; HMO accounts for 73.3% of all plans.

Age-Based Premiums

Reference monthly premiums at age 21 are $320 (Bronze), $430 (Silver), $530 (Gold), and $640 (Platinum), scaled by the MA 2:1 rating curve. Figure 2.3 shows premiums plateau from age 54 onward.

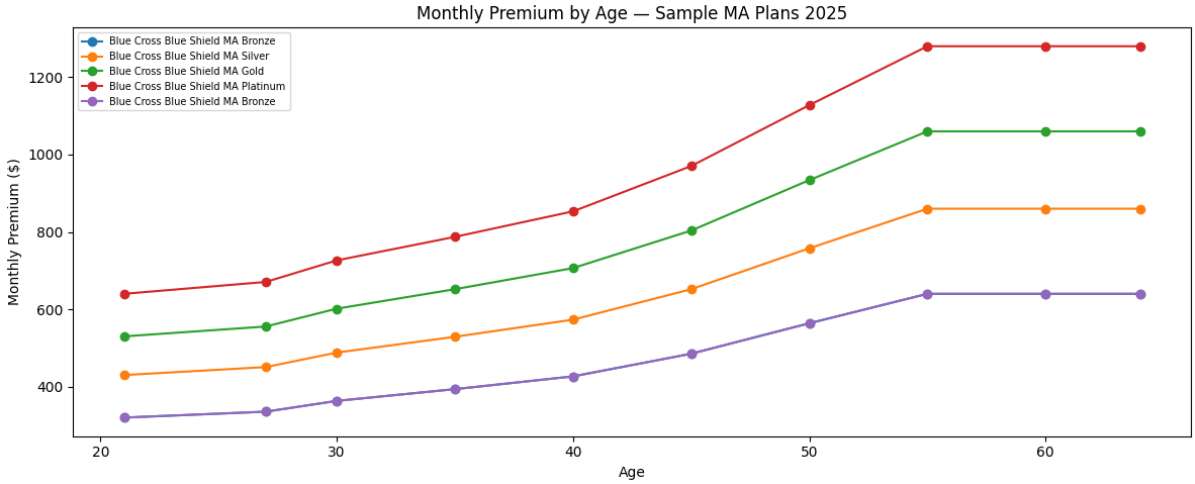

Figure 2.3: Monthly premium by age for BCBS MA plans (Cell 6). Platinum reaches $1,280/month at age 54 and plateaus, reflecting MA's 2:1 cap.

ConnectorCare Subsidy Tiers

Table 2.2 summarizes the 7 ConnectorCare plan types. Figure 2.4 shows the copay heatmap; inpatient hospital ($250) is the highest single copay for Types 3A to 3D, and PCP visits are $0 across all plan types.

Table 2.2: ConnectorCare Plan Types (2025, Table 4)

Type	FPL	Min Premium	Deductible	Med OOP	Specialist
Type 1	0 to 100%	$0	$0	$0	$0
Type 2A	100 to 150%	$0	$0	$750	$18
Type 2B	150 to 200%	$51	$0	$750	$18
Type 3A	200 to 250%	$99	$0	$1,500	$22
Type 3B	250 to 300%	$147	$0	$1,500	$22
Type 3C	300 to 400%	$226	$0	$1,500	$22
Type 3D	400 to 500%	$264	$0	$1,500	$22

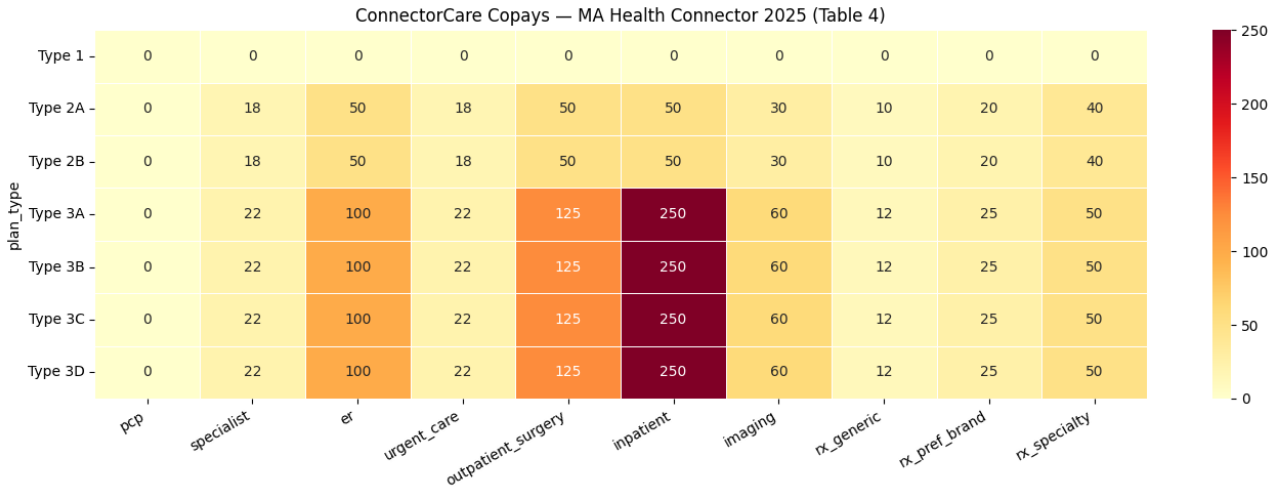

Figure 2.4: ConnectorCare copay heatmap (Cell 6). Type 1 has $0 copays across all services. Inpatient hospital ($250, dark red) is the highest copay for Types 3A to 3D.

Re-Ranker and Query Design

The XGBoost re-ranker uses 11 features listed in Table 2.3. User queries are enriched with filter context before embedding: $q' = [\text{age } A,\ \text{Tier } T,\ \text{Carrier } C] + q$.

Table 2.3: XGBoost Re-Ranker Feature Set

Feature	Description
`faiss_score`	Cosine similarity from FAISS
`keyword_overlap`	Jaccard overlap between query and chunk words
`tier_match`	Metal tier in query matches chunk tier
`carrier_match`	Carrier name in query matches chunk carrier
`type_match`	Plan type (HMO/PPO) match
`is_connectorcare`	Chunk is a ConnectorCare entry
`cc_query`	Query mentions ConnectorCare, FPL, or income
`chunk_length`	Normalized chunk length
`premium_query`	Query mentions premium, cost, or price
`has_age`	Query contains a 2-digit age
`feedback_score`	Historical average feedback (default 0.5)

Project Demonstration

Application Interface

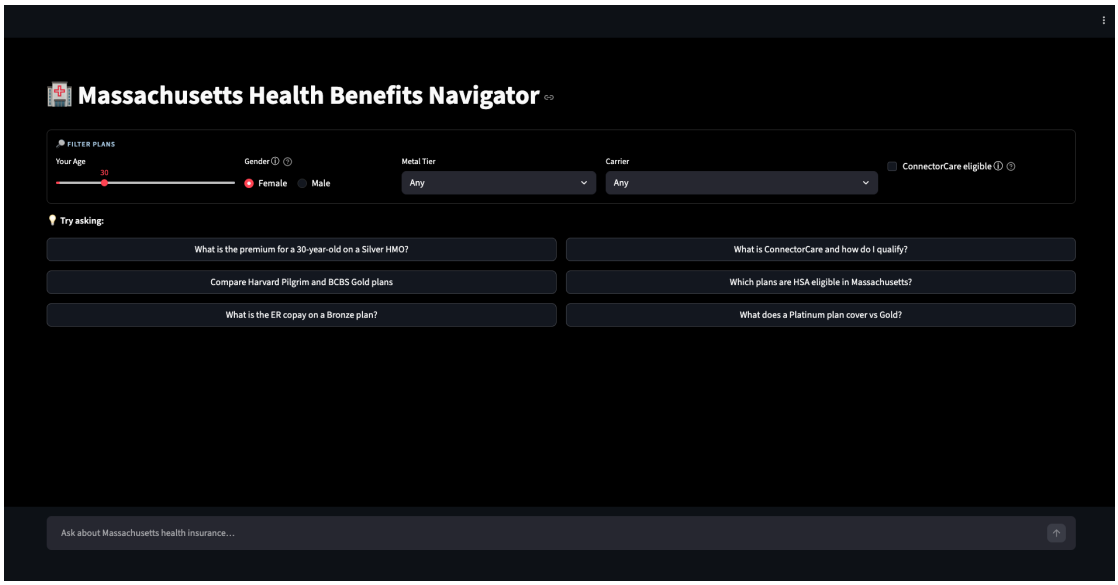

Figure 3.1: Streamlit app home screen (Cell 12). The filter bar supports age (18 to 64), metal tier, carrier, and ConnectorCare eligibility selection. Six pre-populated suggestion buttons are shown on first load.

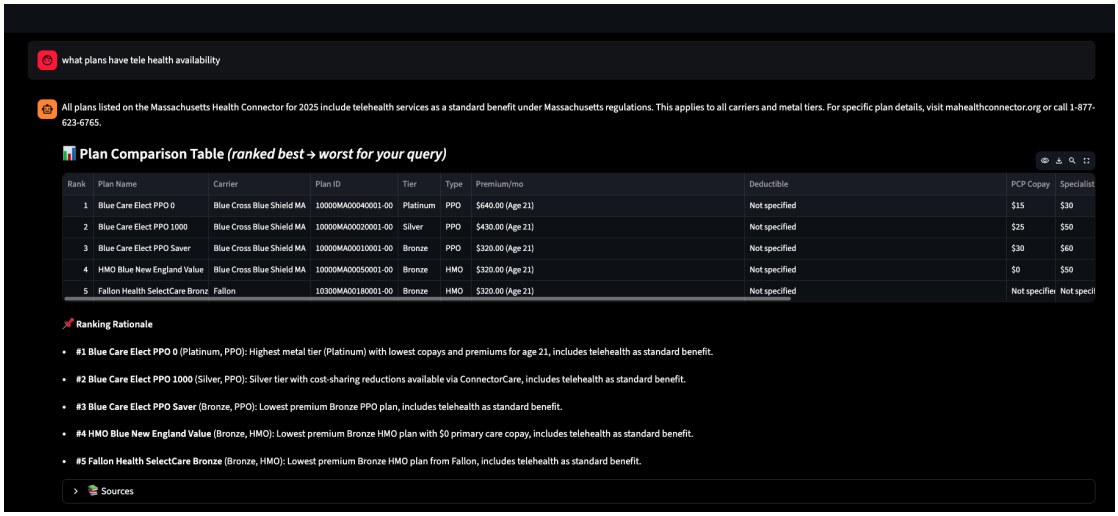

Figure 3.2: Example chat response (Cell 12). Mistral returns a structured JSON rendered as a narrative answer followed by a ranked plan comparison table with per-plan rationale.

Evaluation

Framework

Evaluation uses 36 questions auto-generated by Mistral from plan and ConnectorCare chunks (30 plan questions, 6 ConnectorCare questions). A rate limit during generation reduced the target of 50 questions to 36. LLM quality uses a 20-question sub-sample evaluated by Mistral-as-judge on Faithfulness and Relevancy (0 to 5 scale).

Results

Tables 4.1 to 4.4 report results across all three pipeline stages. Figure 4.1 shows the full six-panel evaluation dashboard from Cell 14.

Table 4.1: Stage 1: FAISS Retrieval (n=36)

Metric	**Score**	**Note**
Hit Rate @15	0.8611	Correct chunk in top 15 for 86% of queries
MRR	0.7265	Relevant chunk near rank 1 on average
Avg. FAISS Score	0.6000	Moderate cosine similarity

Table 4.2: Stage 2: XGBoost Re-Ranker (n=36)

Metric	**Score**	**Note**
NDCG@5	0.4843	Moderate ranking quality
Precision@1	0.4167	Correct chunk at rank 1 in 42% of queries
Precision@3	0.1574	Low
MRR (XGBoost)	0.4593	Lower than raw FAISS
MRR (FAISS only)	0.7245	Baseline
MRR Improvement	−0.2653	Re-ranker hurts ranking vs. FAISS alone

Table 4.3: Stage 3: Mistral LLM Quality (n=20)

Metric	**Score**	**Note**
JSON Validity Rate	1.00 (100%)	All responses parsed as valid JSON
Avg. Faithfulness (/5)	4.15	Strong grounding in retrieved context
Avg. Relevancy (/5)	4.25	Answers address the question well

The overall pipeline score is:

$$\text{Score} = \tfrac{1}{4}(0.861 + 0.484 + 0.830 + 0.850) = \mathbf{0.756}$$

Table 4.4: Overall Pipeline Health

Component	**Metric**	**Score**	**Normalized**
Retrieval	Hit Rate @15	0.8611	0.86
Re-Ranking	NDCG@5	0.4843	0.48
LLM Quality	Faithfulness (/5)	4.15	0.83
LLM Quality	Relevancy (/5)	4.25	0.85
Overall Pipeline Score			**0.756 / 1.0**

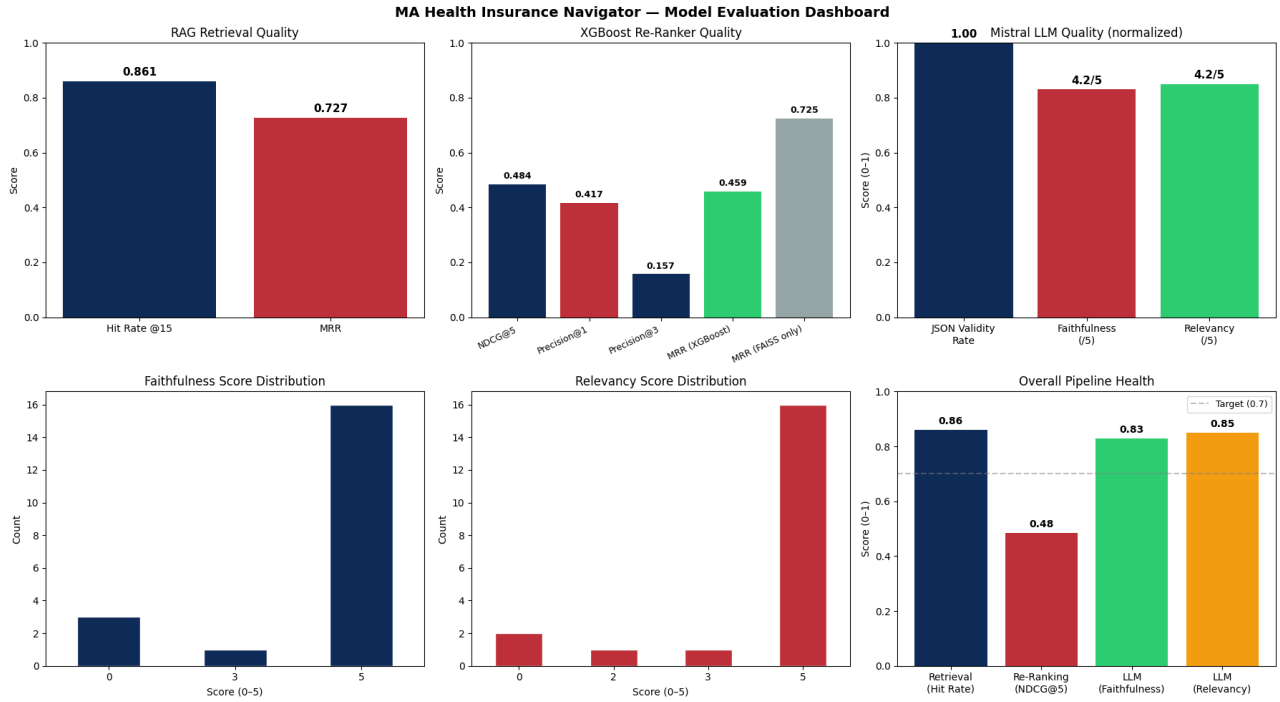

Figure 4.1: Model evaluation dashboard (Cell 14). Top row: FAISS retrieval (Hit Rate 0.861, MRR 0.727), XGBoost re-ranker (NDCG@5 0.484, MRR improvement -0.265), and Mistral LLM quality (100% JSON validity, Faithfulness 4.2/5, Relevancy 4.2/5). Bottom row: score distributions (16/20 answers scored 5/5 on both axes) and overall pipeline health vs. the 0.70 target threshold.

Discussion and Conclusion

Discussion

Retrieval performs strongly (Hit Rate 0.861, MRR 0.727). The structured plain-text chunk format, which encodes carrier, tier, and copay fields as human-readable text, aligns well with natural language queries and produces high cosine similarity for on-topic questions.

Re-ranking is the weakest pipeline stage. The -0.265 MRR degradation is attributable to the extremely small training set of 15 synthetic query-label pairs. The dominant signal in the feature set, `faiss_score`, is already the most informative feature for ranking; the re-ranker introduces noise rather than signal and should be replaced with a cross-encoder or retrained on substantially more labeled data.

LLM generation is the strongest stage (Faithfulness 4.15/5, Relevancy 4.25/5, 100% JSON validity). The strict output schema and `temperature=0.1` constrain hallucination risk. The 3 responses scoring 0/5 on Faithfulness are consistent with the API rate-limit errors observed during the evaluation run.

Limitations include: (1) re-ranker trained on only 15 examples; (2) test set reduced from 50 to 36 due to rate limiting; (3) no provider network data; (4) static 2025 plan data requiring annual rebuilds; and (5) Mistral API key embedded in the app file rather than loaded from a secrets manager.

Future Work

Priority improvements are: replacing XGBoost with a fine-tuned cross-encoder (e.g., `ms-marco-MiniLM-L-6-v2`); collecting real user queries from the feedback log to build a supervised re-ranker training set; integrating MA DOI provider network data for doctor-specific queries; adding multi-turn dialogue support; and automating annual CMS PUF refreshes.

The most significant long-term opportunity is **national expansion**. The CMS Public Use Files are published for all 50 states and the federal exchange, covering every ACA-compliant plan in the country. Because the system is built entirely on the standardized CMS PUF schema, extending coverage to additional states requires no architectural changes only state-specific data ingestion, chunk construction, and FAISS index builds. A 50-state deployment would make the navigator the first conversational, RAG-powered health insurance tool covering the full US individual market, serving tens of millions of consumers navigating open enrollment across all state-based and federally facilitated exchanges.

Conclusion

The Massachusetts Health Insurance Navigator achieves an overall pipeline score of 0.756/1.0, exceeding the 0.70 production threshold, driven by strong FAISS retrieval (0.861 Hit Rate) and high-quality Mistral generation (4.15/5 Faithfulness). The system demonstrates that domain-constrained RAG pipelines are viable for structured, tabular knowledge bases in regulated consumer information domains. The primary finding for future work is that the XGBoost re-ranker requires substantially more training data or architectural replacement to add value over raw semantic retrieval.

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